Organisms and their environment

Mark scheme and worked solutions · 30 questions

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Question 1

core · 1 mark · 0610_T19_P1_20261018_Q1

Answer: B

Rationale
Sunlight supplies the principal energy input, captured by producers. Minerals and carbon dioxide supply materials rather than the main energy input.

Question 2

core · 1 mark · 0610_T19_P1_20261018_Q2

Answer: D

Rationale
A producer makes organic nutrients, usually through photosynthesis. Rabbits and hawks consume organisms; the fungus is a decomposer.

Question 3

core · 1 mark · 0610_T19_P1_20261018_Q3

Answer: A

Rationale
The grasshopper is a primary consumer feeding on the producer. The lizard eats that primary consumer and is therefore secondary.

Question 4

core · 1 mark · 0610_T19_P1_20261018_Q4

Answer: C

Rationale
Decomposers obtain energy from dead or waste organic material. Photosynthesis identifies producers and grazing identifies herbivores.

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Question 5

core · 1 mark · 0610_T19_P1_20261018_Q5

Answer: D

Rationale
Carnivores obtain energy by eating animals. Herbivores eat plants; producers make their own organic nutrients.

Question 6

core · 1 mark · 0610_T19_P1_20261018_Q6

Answer: B

Rationale
A food web links several chains and can show multiple food sources. It includes producers and does not imply energy recycling.

Question 7

core · 1 mark · 0610_T19_P1_20261018_Q7

Answer: C

Rationale
Removing predators reduces mortality of their small-fish prey, allowing those fish to increase. The other outcomes do not follow from the given relationships.

Question 8

core · 1 mark · 0610_T19_P1_20261018_Q8

Answer: A

Rationale
Biomass takes the amount of living material into account. Counts treat a tree and a small herbivore as one each; biomass is not itself an energy-flow rate.

Question 9

supplement · 1 mark · 0610_T19_P1_20261018_Q9

Answer: B

Rationale
Consumers respire and energy is eventually transferred as heat. Animal muscle cells do not photosynthesise, and energy is not all returned to producers.

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Question 10

supplement · 1 mark · 0610_T19_P1_20261018_Q10

Answer: C

Rationale
Energy transfer is inefficient, leaving less available at higher levels. This limits the populations that a higher trophic level can support.

Question 11

supplement · 1 mark · 0610_T19_P1_20261018_Q11

Answer: D

Rationale
Cattle use energy in respiration and do not convert all crop material into edible tissue. Direct consumption avoids an additional transfer step.

Question 12

supplement · 1 mark · 0610_T19_P1_20261018_Q12

Answer: A

Rationale
Nitrogen fixation converts atmospheric nitrogen into compounds. Denitrification releases nitrogen gas, while evaporation and carbohydrate combustion do not fix it.

Question 13

core · 1 mark · 0610_T19_P1_20261018_Q13

Answer: C

Rationale
A population contains organisms of one species in the same area at the same time. A community includes different species.

Question 14

core · 1 mark · 0610_T19_P1_20261018_Q14

Answer: A

Rationale
A community includes all the populations of different species. One species is a population; physical conditions alone are not a community.

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Question 15

core · 1 mark · 0610_T19_P1_20261018_Q15

Answer: B

Rationale
Disease can increase mortality or reduce reproduction and therefore limit population growth. It does not create resources or make animals producers.

Question 16

core · 4 marks · 0610_T19_P1_20261018_Q16

(a) [2]

- Green plants absorb light energy from the Sun. (1)
- Photosynthesis stores energy as chemical energy in organic nutrients. (1)

(b) [2]

- Feeding transfers chemical energy between organisms. (1)
- Energy is eventually transferred to the environment, for example as heat from respiration. (1)

Worked solution
(a) Green plants absorb light energy from the Sun. Photosynthesis stores energy as chemical energy in organic nutrients.

(b) Feeding transfers chemical energy between organisms. Energy is eventually transferred to the environment, for example as heat from respiration.

Question 17

core · 4 marks · 0610_T19_P1_20261018_Q17

(a) [2]

- Pondweed → snail. (1)
- Snail → fish → heron, completing the chain in this order. (1)

(b) [2]

- Pondweed is the producer. (1)
- The heron is the tertiary consumer. (1)

Worked solution
(a) Pondweed → snail. Snail → fish → heron, completing the chain in this order.

(b) Pondweed is the producer. The heron is the tertiary consumer.

Question 18

core · 4 marks · 0610_T19_P1_20261018_Q18

(a) [2]

- Owls have less available food. (1)

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- Owl survival or reproduction may decrease, reducing their population. (1)

(b) [2]

- Foxes can also eat rabbits. (1)
- This alternative food source reduces their dependence on voles. (1)

Worked solution
(a) Owls have less available food. Owl survival or reproduction may decrease, reducing their population.

(b) Foxes can also eat rabbits. This alternative food source reduces their dependence on voles.

Question 19

core · 4 marks · 0610_T19_P1_20261018_Q19

(a) [2]

- It obtains energy by feeding on other organisms. (1)
- Those organisms are plants, so it is a herbivore. (1)

(b) [2]

- It obtains energy by eating animals. (1)
- It is a secondary consumer in the leaf → hare → wildcat chain. (1)

Worked solution
(a) It obtains energy by feeding on other organisms. Those organisms are plants, so it is a herbivore.

(b) It obtains energy by eating animals. It is a secondary consumer in the leaf → hare → wildcat chain.

Question 20

core · 4 marks · 0610_T19_P1_20261018_Q20

(a) [2]

- They obtain energy from dead organic material. (1)
- They are decomposers. (1)

(b) [2]

- Plants make organic nutrients through photosynthesis. (1)
- They use light energy rather than obtaining all energy by consuming dead material. (1)

Worked solution
(a) They obtain energy from dead organic material. They are decomposers.

(b) Plants make organic nutrients through photosynthesis. They use light energy rather than obtaining all energy by consuming dead material.

Question 21

core · 4 marks · 0610_T19_P1_20261018_Q21

(a) [2]

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- Counting gives each tree and each insect equal weight. (1)
- It ignores the much greater amount of living material in one tree. (1)

(b) [2]

- Biomass accounts for differences in organism size. (1)
- It better represents the amount of material available at each trophic level. (1)

Worked solution
(a) Counting gives each tree and each insect equal weight. It ignores the much greater amount of living material in one tree.

(b) Biomass accounts for differences in organism size. It better represents the amount of material available at each trophic level.

Question 22

supplement · 4 marks · 0610_T19_P1_20261018_Q22

(a) [2]

- It represents the energy available to support organisms at each level. (1)
- It measures transfer over time rather than just a snapshot of stored biomass. (1)

(b) [2]

- Some energy is used in respiration and transferred as heat. (1)
- Some material is not eaten or is not digested and absorbed. (1)

Worked solution
(a) It represents the energy available to support organisms at each level. It measures transfer over time rather than just a snapshot of stored biomass.

(b) Some energy is used in respiration and transferred as heat. Some material is not eaten or is not digested and absorbed.

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Question 23

supplement · 4 marks · 0610_T19_P1_20261018_Q23

(a) [2]

- 2700 divided by 18 000 gives 0.15. (1)
- Multiplying by 100 gives 15%. (1)

(b) [2]

- Not all plant material is eaten or digested. (1)
- Herbivores use absorbed energy in respiration, transferring energy to the environment. (1)

Worked solution
(a) 2700 divided by 18 000 gives 0.15. Multiplying by 100 gives 15%.

(b) Not all plant material is eaten or digested. Herbivores use absorbed energy in respiration, transferring energy to the environment.

Question 24

supplement · 4 marks · 0610_T19_P1_20261018_Q24

(a) [2]

- It has fewer trophic transfers. (1)
- It avoids energy used by livestock in respiration and lost in uneaten or waste material. (1)

(b) [2]

- Each transfer passes on only a fraction of the available energy. (1)
- Eventually too little energy remains to support a further substantial consumer population. (1)

Worked solution

(a) It has fewer trophic transfers. It avoids energy used by livestock in respiration and lost in uneaten or waste material.

(b) Each transfer passes on only a fraction of the available energy. Eventually too little energy remains to support a further substantial consumer population.

Question 25

core · 4 marks · 0610_T19_P1_20261018_Q25

(a) [2]

- The introduced insect reduces the birds' food supply. (1)
- Increased competition may reduce bird survival and reproduction. (1)

(b) [2]

- Fewer birds provide less food for their predator. (1)
- The predator population may then decrease or switch to other prey. (1)

Worked solution

(a) The introduced insect reduces the birds' food supply. Increased competition may reduce bird survival and reproduction.

(b) Fewer birds provide less food for their predator. The predator population may then decrease or switch to other prey.

Question 26

core · 4 marks · 0610_T19_P1_20261018_Q26

(a) [2]

- Plants absorb carbon dioxide during photosynthesis. (1)
- Herbivores obtain carbon compounds by feeding on plants. (1)

(b) [2]

- Respiration by plants, animals or decomposers. (1)
- Combustion of wood or fossil fuels. (1)

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Worked solution
(a) Plants absorb carbon dioxide during photosynthesis. Herbivores obtain carbon compounds by feeding on plants.

(b) Respiration by plants, animals or decomposers. Combustion of wood or fossil fuels.

Question 27

supplement · 4 marks · 0610_T19_P1_20261018_Q27

(a) [2]

- Decomposers convert nitrogen in proteins to ammonium ions. (1)
- Nitrifying bacteria convert ammonium compounds to nitrate ions that plants can absorb. (1)

(b) [2]

- Nitrogen fixation converts nitrogen gas to nitrogen compounds. (1)
- Denitrification converts nitrate ions to nitrogen gas, returning it to the atmosphere. (1)

Worked solution
(a) Decomposers convert nitrogen in proteins to ammonium ions. Nitrifying bacteria convert ammonium compounds to nitrate ions that plants can absorb.

(b) Nitrogen fixation converts nitrogen gas to nitrogen compounds. Denitrification converts nitrate ions to nitrogen gas, returning it to the atmosphere.

Question 28

core · 4 marks · 0610_T19_P1_20261018_Q28

(a) [2]

- The reed population is the organisms of that one species in the area at the same time. (1)
- The community contains all populations of different species in the wetland. (1)

(b) [2]

- It includes a community and its physical environment. (1)
- The organisms and the environment interact with one another. (1)

Worked solution
(a) The reed population is the organisms of that one species in the area at the same time. The community contains all populations of different species in the wetland.

(b) It includes a community and its physical environment. The organisms and the environment interact with one another.

Question 29

core · 4 marks · 0610_T19_P1_20261018_Q29

(a) [2]

- Rabbits compete more for limited food. (1)
- Less food can reduce survival or reproduction. (1)

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(b) [2]

- More foxes can kill more rabbits. (1)
- This increases mortality and reduces net population growth. (1)

Worked solution
(a) Rabbits compete more for limited food. Less food can reduce survival or reproduction.

(b) More foxes can kill more rabbits. This increases mortality and reduces net population growth.

Question 30

supplement · 4 marks · 0610_T19_P1_20261018_Q30

(a) [2]

- Initially the bacteria adapt to the conditions and there are few reproducing cells. (1)
- Later many bacteria reproduce while nutrients and other resources are plentiful. (1)

(b) [2]

- Nutrients become limiting and wastes accumulate. (1)
- Birth and death rates become approximately equal in the stationary phase. (1)

Worked solution
(a) Initially the bacteria adapt to the conditions and there are few reproducing cells. Later many bacteria reproduce while nutrients and other resources are plentiful.

(b) Nutrients become limiting and wastes accumulate. Birth and death rates become approximately equal in the stationary phase.

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